Math 105: Calculus with Algebra
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quiz archive

Gradescope code: B5J8EN

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Math 105 Fall 2026 - Take-Home Quiz 1 Solutions
Due Sunday 9/6/2026 - at 11:55 PM

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(1) Solve the equation $|x-2|=7$ for x.

Solution. Since $|x-2|=7$, either $x-2=7$ or $x-2=-7$. Thus $x=9$ or $x=-5$.

(2) Solve the inequality $|x+3|<4$.

Solution. The inequality $|x+3|<4$ is equivalent to $-4<x+3<4$. Subtracting 3 throughout gives $-7<x<1$.

Thus the solution is $(-7,1)$. On the number line, shade the interval between -7 and 1, with open circles at -7 and 1.

(3) Express

$$1+\frac{1}{\frac{x}{x+2}}$$

as a single fraction.

Solution. We have

$$1+\frac{1}{\frac{x}{x+2}}=1+\frac{x+2}{x}=\frac{x+x+2}{x}=\frac{2x+2}{x}.$$

Thus the single fraction is $\dfrac{2x+2}{x}$.

The original expression is defined only when $x\neq -2$ and $x\neq 0$.

(1) Consider the line

$$x + 2y = 2.$$

(a) To find the slope, solve the equation for y:

$$x + 2y = 2$$

$$2y = -x + 2$$

$$y = -\frac{1}{2}x + 1.$$

Thus the slope is

$$\boxed{-\frac{1}{2}}.$$

(b) A line parallel to this line has the same slope, so its slope is $-\frac{1}{2}$. Since it passes through $(2, -1)$, we can use point-slope form:

$$y - (-1) = -\frac{1}{2}(x - 2).$$

Thus

$$y + 1 = -\frac{1}{2}x + 1,$$

and hence

$$\boxed{y = -\frac{1}{2}x}.$$

(c) A perpendicular line has slope equal to the negative reciprocal of the original slope. Since the original slope is $-\frac{1}{2}$, the perpendicular slope is

$$\boxed{2}.$$

(2) Let

$$f(x) = 3x^2 + 2.$$

We want to compute

$$\frac{f(a+h) - f(a)}{h}.$$

First,

$$f(a+h) = 3(a+h)^2 + 2.$$

Expanding gives

$$f(a+h) = 3(a^2 + 2ah + h^2) + 2 = 3a^2 + 6ah + 3h^2 + 2.$$

Also,

$$f(a) = 3a^2 + 2.$$

Therefore

$$\begin{aligned}\frac{f(a+h) - f(a)}{h} &= \frac{3a^2 + 6ah + 3h^2 + 2 - (3a^2 + 2)}{h} \\ &= \frac{6ah + 3h^2}{h} \\ &= \boxed{6a + 3h}.\end{aligned}$$

(3) Let

$$g(x) = \begin{cases} x^2, & \text{if } x \leq 4, \\ -|x - 4|, & \text{if } x > 4. \end{cases}$$

(a) To compute $g(0)$, notice that $0 \leq 4$, so we use the first part of the definition:

$$g(0) = 0^2 = \boxed{0}.$$

Since $5 > 4$, we use the second part to compute $g(5)$:

$$g(5) = -|5 - 4| = -|1| = \boxed{-1}.$$

(b) For $x \leq 4$, the graph is the parabola

$$y = x^2.$$

Since $x = 4$ is included, there is a closed point at

$$(4, 16).$$

For $x > 4$, we have $x - 4 > 0$, so

$$|x - 4| = x - 4.$$

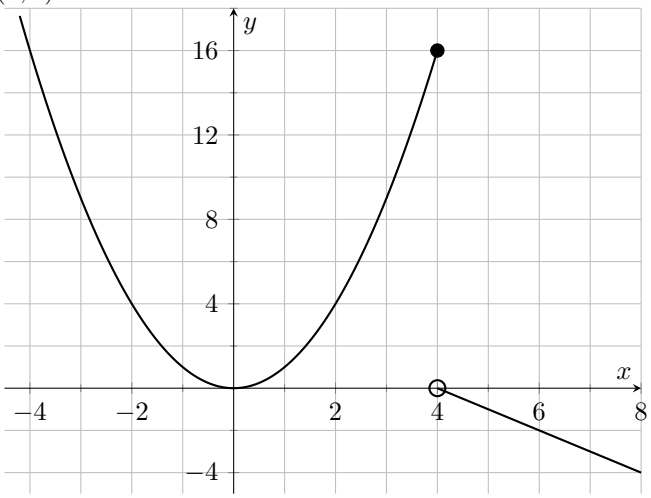
Therefore

$$-|x - 4| = -(x - 4) = 4 - x.$$

Thus the second piece is the line

$$y = 4 - x, \qquad x > 4,$$

with an open point at $(4, 0)$.



The first piece covers all $x \leq 4$ and the second covers all $x > 4$. Therefore the domain is

$$\boxed{(-\infty, \infty)}.$$

The parabola gives every $y \geq 0$, while the line gives every $y < 0$. Together these give every real number. Therefore the range is

$$\boxed{(-\infty, \infty)}.$$

Math 105 Fall 2026 - Take-Home Quiz 3 Solutions

(1) The graph of $y = f(x)$ is shown below, and we want to sketch

$$y = f(x-1) + 2.$$

The expression $f(x-1)$ shifts the graph of f **one unit to the right**. Then adding 2 shifts the graph **two units up**.

Thus every point

$$(x, y)$$

on the original graph moves to

$$(x+1, y+2).$$

In particular,

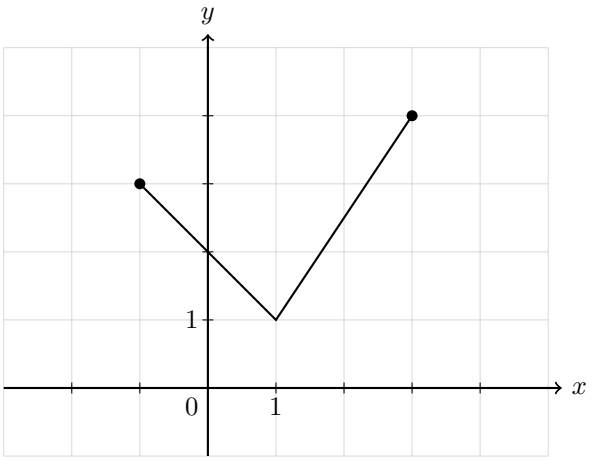
$$(-2, 1) \longmapsto (-1, 3),$$

$$(0, -1) \longmapsto (1, 1),$$

and

$$(2, 2) \longmapsto (3, 4).$$

Therefore the transformed graph is:



Shift right 1 unit and up 2 units.

(2) Let

$$f(x) = \sqrt{x+1} \qquad \text{and} \qquad g(x) = x^2 - 2.$$

We want to compute

$$(f \circ g)(2).$$

Recall that

$$(f \circ g)(2) = f(g(2)).$$

First compute the inside function:

$$g(2) = 2^2 - 2 = 4 - 2 = 2.$$

Then

$$f(g(2)) = f(2) = \sqrt{2+1} = \sqrt{3}.$$

Therefore

$$\boxed{(f \circ g)(2) = \sqrt{3}}.$$

(3) Find

$$\lim_{x\to 3} \frac{x^2-9}{x-3}.$$

If we substitute $x = 3$ immediately, we get

$$\frac{3^2-9}{3-3} = \frac{0}{0},$$

so we need to simplify first.

Factor the numerator:

$$x^2 - 9 = (x-3)(x+3).$$

Therefore, for $x \neq 3$,

$$\frac{x^2-9}{x-3} = \frac{(x-3)(x+3)}{x-3} = x+3.$$

Now we can take the limit:

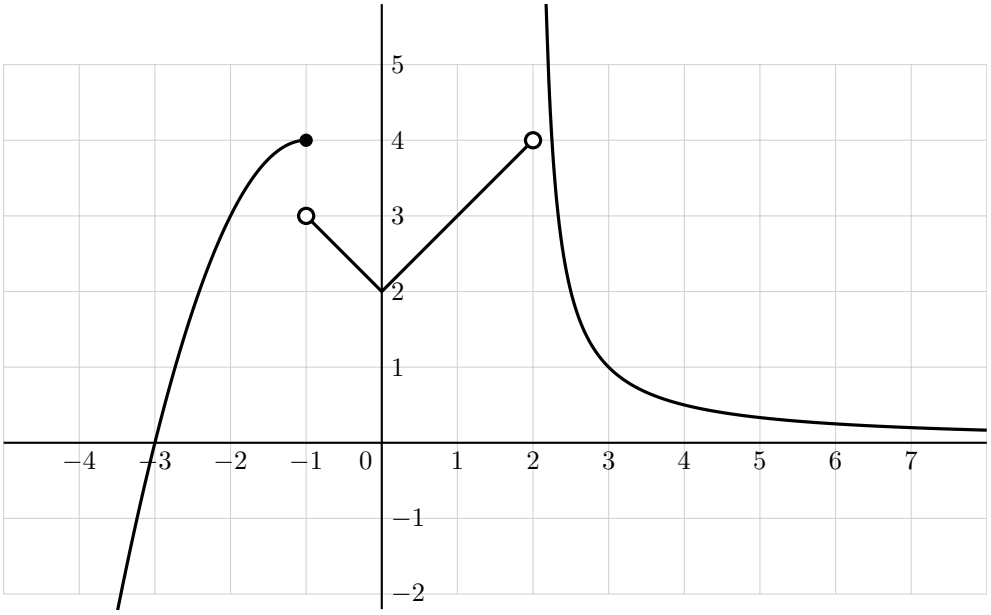
$$\lim_{x\to 3}(x+3) = 3+3 = 6.$$

Thus

$$\boxed{\lim_{x\to 3} \frac{x^2-9}{x-3} = 6}.$$

Math 105 Fall 2026 - Take-Home Quiz 4 Solutions

(1) (5 points) Consider the following graph of $f(x)$.



(a) As x approaches 2 from the right, the graph rises without bound. Therefore

$$\boxed{\lim_{x\to 2^+} f(x) = +\infty.}$$

(b) As x approaches 2 from the left, the graph approaches the open point $(2, 4)$. Therefore

$$\boxed{\lim_{x\to 2^-} f(x) = 4.}$$

(c) The right-hand limit is $+\infty$, while the left-hand limit is 4. Since the one-sided limits do not agree,

$$\boxed{\lim_{x\to 2} f(x) \text{ does not exist.}}$$

(d) As $x \to -1^-$, the graph approaches 4, while as $x \to -1^+$, the graph approaches 3. Since the one-sided limits are different,

$$\boxed{\lim_{x\to -1} f(x) \text{ does not exist.}}$$

(e) Near $x = 0$, the graph lies on the two line segments meeting at $(0, 2)$, so from both sides $f(x)$ approaches 2. Thus

$$\boxed{\lim_{x\to 0} f(x) = 2.}$$

(2) (1 point)

$$\lim_{x\to -3} \frac{x+3}{x^2+x+1}.$$

The denominator is nonzero at $x=-3$, since $(-3)^2+(-3)+1=7$. Therefore direct substitution gives

$$\boxed{0}.$$

(3) (1 point)

$$\lim_{x\to 3} \frac{x-7}{3-x}.$$

As $x\to 3$, the numerator approaches -4. If $x\to 3^-$, then $3-x\to 0^+$, so the quotient approaches $-\infty$. If $x\to 3^+$, then $3-x\to 0^-$, so the quotient approaches $+\infty$. Therefore

$$\boxed{\text{the limit does not exist.}}$$

(4) (1 point)

$$\lim_{x\to -3} \frac{x+3}{x^2-2x-15}.$$

Factor the denominator:

$$x^2-2x-15=(x-5)(x+3).$$

For $x\neq -3$,

$$\frac{x+3}{(x-5)(x+3)}=\frac{1}{x-5}.$$

Therefore

$$\boxed{-\frac{1}{8}}.$$

(5) (2 points)

$$\lim_{x\to -3} \frac{\dfrac{1}{1-x}-\dfrac{1}{4}}{x+3}.$$

Combine the fractions in the numerator:

$$\frac{1}{1-x}-\frac{1}{4}=\frac{4-(1-x)}{4(1-x)}=\frac{x+3}{4(1-x)}.$$

Hence, for $x\neq -3$,

$$\frac{\dfrac{x+3}{4(1-x)}}{x+3}=\frac{1}{4(1-x)}.$$

Now substitute $x=-3$:

$$\boxed{\frac{1}{16}}.$$

Math 105: Midterm 1 Solution Guide

1. Consider the line L_1 given by $3x + 4y = 5$.

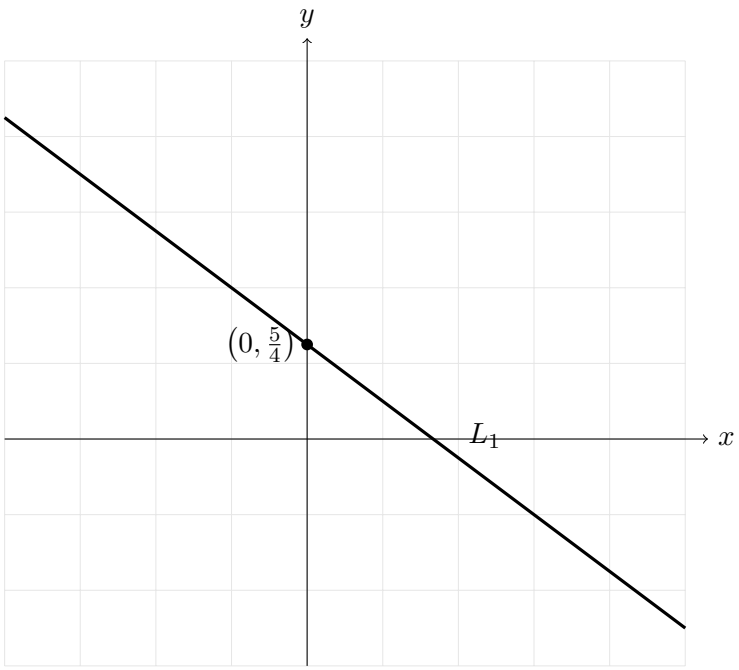
   (a) Solve for y:

   $$3x + 4y = 5 \qquad \Longrightarrow \qquad y = -\frac{3}{4}x + \frac{5}{4}.$$

   Thus L_1 has slope $-\frac{3}{4}$ and y-intercept

   $$\boxed{\left(0, \frac{5}{4}\right)}.$$

   A convenient second point is the x-intercept $\left(\frac{5}{3}, 0\right)$.



   (b) A line perpendicular to L_1 has slope equal to the opposite reciprocal of $-\frac{3}{4}$, namely $\frac{4}{3}$. Using the point $(-5, -1)$,

   $$y + 1 = \frac{4}{3}(x + 5).$$

   Solving for y gives

   $$\boxed{y = \frac{4}{3}x + \frac{17}{3}}.$$

   (c) A line parallel to L_1 must have the same slope. Therefore

   $$\boxed{m = -\frac{3}{4}}.$$

2. Let $f(x) = x^2 - x$ and $g(x) = \sqrt{x + 4}$.

(a) Substitute $2x^3$ for x in f:

$$f(2x^3) = (2x^3)^2 - 2x^3 = \boxed{4x^6 - 2x^3}.$$

(b) Substitute x^2 for x in g:

$$g(x^2) = \boxed{\sqrt{x^2+4}}.$$

(c)

$$(f \circ g)(x) = f(g(x)) = \left(\sqrt{x+4}\right)^2 - \sqrt{x+4}.$$

Therefore

$$\boxed{(f \circ g)(x) = x + 4 - \sqrt{x+4}}.$$

The expression is defined when $x + 4 \geq 0$, so its domain is $[-4, \infty)$.

3. Evaluate each limit.

(a)

$$\lim_{x \to 4} \frac{x^2 - x - 12}{x^2 - 3x - 4}.$$

Direct substitution gives $0/0$, so factor:

$$\frac{(x-4)(x+3)}{(x-4)(x+1)} = \frac{x+3}{x+1} \qquad (x \neq 4).$$

Thus

$$\boxed{\frac{7}{5}}.$$

(b)

$$\lim_{x \to -1} \frac{x^2 - 1}{2 - \sqrt{x+5}}.$$

Again direct substitution gives $0/0$. Rationalize the denominator:

$$\frac{x^2-1}{2-\sqrt{x+5}} \cdot \frac{2+\sqrt{x+5}}{2+\sqrt{x+5}} = \frac{(x-1)(x+1)(2+\sqrt{x+5})}{4-(x+5)}.$$

Since $4 - (x+5) = -(x+1)$, this simplifies to

$$-(x-1)(2+\sqrt{x+5}).$$

Now substitute $x = -1$:

$$-(-2)(2+2) = \boxed{8}.$$

(c)

$$\lim_{x \to 2} \frac{x^2 - 4}{4 - x}.$$

The denominator is nonzero at $x = 2$, so direct substitution works:

$$\frac{4-4}{4-2} = \boxed{0}.$$

(d)

$$\lim_{x\to 2} \frac{x+1}{(x-2)^2}.$$

The numerator approaches $3 > 0$, while $(x-2)^2 \to 0^+$ from both sides. Therefore

$$\boxed{+\infty}.$$

(e)

$$\lim_{x\to 1} \frac{x^2-3x+2}{|x-1|}.$$

Factor the numerator:

$$\frac{(x-1)(x-2)}{|x-1|}.$$

For $x > 1$, $|x-1| = x-1$, so

$$\lim_{x\to 1^+} \frac{x^2-3x+2}{|x-1|} = \lim_{x\to 1^+} (x-2) = -1.$$

For $x < 1$, $|x-1| = -(x-1)$, so

$$\lim_{x\to 1^-} \frac{x^2-3x+2}{|x-1|} = \lim_{x\to 1^-} -(x-2) = 1.$$

The one-sided limits are different, so

$$\boxed{\text{DNE}}.$$

4. Sketch $f(x) = 1 + \dfrac{2}{x-1}$.

The denominator is zero at $x = 1$, so the vertical asymptote is

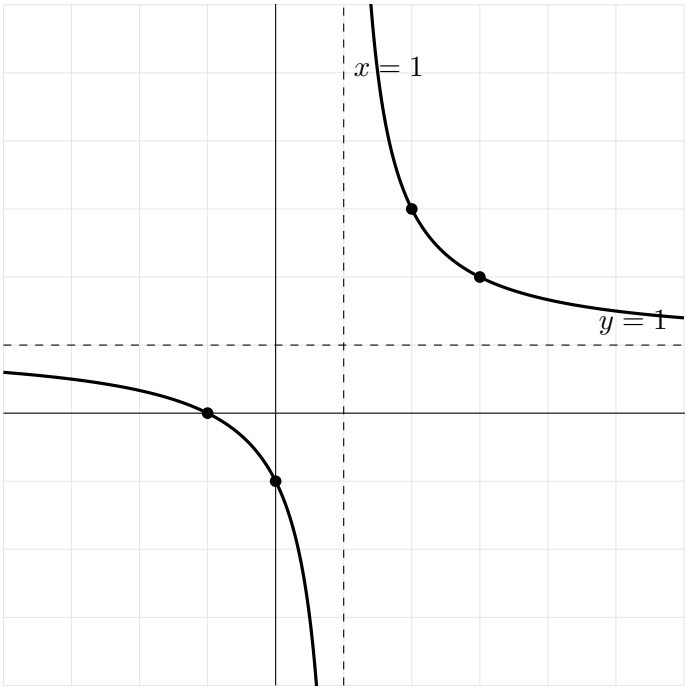
$$\boxed{x = 1}.$$

As x becomes very large in magnitude, $\frac{2}{x-1} \to 0$, so the horizontal asymptote is

$$\boxed{y = 1}.$$

Some convenient points are

$$f(-1) = 0, \qquad f(0) = -1, \qquad f(2) = 3, \qquad f(3) = 2.$$



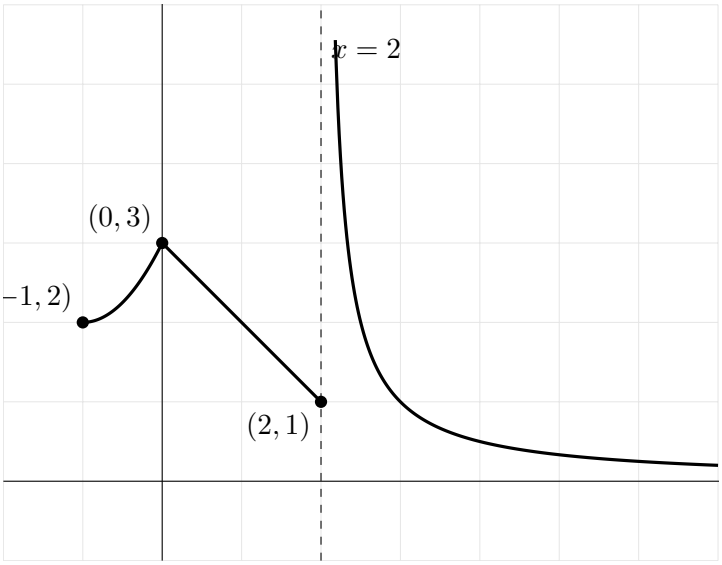
5. Let $f(x) = \dfrac{1}{x}$.

   (a)
   $$\frac{f(x+h) - f(x)}{h} = \frac{\frac{1}{x+h} - \frac{1}{x}}{h}.$$
   Combine the fractions in the numerator:
   $$\frac{1}{x+h} - \frac{1}{x} = \frac{x - (x+h)}{x(x+h)} = \frac{-h}{x(x+h)}.$$
   Therefore
   $$\frac{f(x+h) - f(x)}{h} = \boxed{-\frac{1}{x(x+h)}}.$$

   (b) Using the simplified expression from part (a),
   $$\lim_{h \to 0} -\frac{1}{x(x+h)} = \boxed{-\frac{1}{x^2}},$$
   for $x \neq 0$.

6. Consider
   $$f(x) = \begin{cases} \dfrac{1}{x-2}, & x > 2, \\ 3 - |x|, & 0 < x \leq 2, \\ (x+1)^2 + 2, & -1 \leq x \leq 0. \end{cases}$$

   (a) Since $0 < x \leq 2$ in the middle piece, $|x| = x$, so that piece is the line $y = 3 - x$. A sketch is shown below.



(b) Since the middle piece includes $x = 2$,

$$f(2) = 3 - |2| = \boxed{1}.$$

(c) The three pieces cover every input from -1 onward, so

$$\boxed{\text{Domain} = [-1, \infty)}.$$

For $x > 2$, the function $1/(x - 2)$ takes every positive value. Therefore

$$\boxed{\text{Range} = (0, \infty)}.$$

(d) At $x = 0$,

$$\lim_{x \to 0^-} f(x) = 3 \qquad \text{and} \qquad \lim_{x \to 0^+} f(x) = 3.$$

Hence

$$\boxed{\lim_{x \to 0} f(x) = 3}.$$

At $x = 2$,

$$\lim_{x \to 2^-} f(x) = 1, \qquad \lim_{x \to 2^+} f(x) = +\infty.$$

The one-sided limits do not agree, so

$$\boxed{\lim_{x \to 2} f(x) = \text{DNE}}.$$

Math 105 Fall 2026 - Take-Home Quiz 5 Solutions
Due Sunday 10/18/2026 - at 11:55 PM

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(1) Let

$$f(x) = |x|.$$

Using the limit definition,

$$f'(x) = \lim_{h\to 0} \frac{|x+h| - |x|}{h}.$$

Suppose first that $x > 0$. For h sufficiently close to 0, we also have $x + h > 0$, so

$$\frac{|x+h| - |x|}{h} = \frac{(x+h) - x}{h} = 1.$$

Therefore

$$f'(x) = 1 \qquad \text{when } x > 0.$$

Now suppose that $x < 0$. For h sufficiently close to 0, we also have $x + h < 0$, so

$$\frac{|x+h| - |x|}{h} = \frac{-(x+h) - (-x)}{h} = \frac{-h}{h} = -1.$$

Therefore

$$f'(x) = -1 \qquad \text{when } x < 0.$$

At $x = 0$,

$$f'(0) = \lim_{h\to 0} \frac{|h|}{h}.$$

If $h > 0$, then $|h|/h = 1$, while if $h < 0$, then $|h|/h = -1$. The one-sided limits are different, so $f'(0)$ does not exist.

Thus

$$\boxed{f'(x) = \begin{cases} -1, & x < 0, \\ \text{does not exist}, & x = 0, \\ 1, & x > 0. \end{cases}}$$

(2) The matches are

$$\boxed{A \to 2, \qquad B \to 3, \qquad C \to 4, \qquad D \to 1.}$$

- **Function A** is a smooth cubic. Its derivative is the upward-opening parabola in graph 2. Since A is a polynomial, it is differentiable for every real number.
- **Function B** is a smooth parabola with a minimum at $x = 1$. Its derivative is the increasing line in graph 3, which is 0 at $x = 1$. Thus B is differentiable for every real number.
- **Function C** is a V-shaped graph with a corner at $x = 1$. Its slope is $-\frac{1}{2}$ to the left of 1 and $\frac{1}{2}$ to the right, so its derivative is graph 4. It is not differentiable at

$$\boxed{x = 1}.$$

- **Function D** has the same basic V-shape as C, but is steeper. Its slope is -1 to the left of 1 and 1 to the right, so its derivative is graph 1. It is not differentiable at

$$\boxed{x = 1}.$$

Thus the differentiability table is

Function	Where is it not differentiable?
A	nowhere
B	nowhere
C	$x = 1$
D	$x = 1$

(3) Use the power rule (and the sum rule) to compute the derivatives.

(a)

$$f(x) = 4x^5 - 3x^2 + 7x - 9.$$

Differentiate term by term:

$$f'(x) = 20x^4 - 6x + 7.$$

Therefore

$$\boxed{f'(x) = 20x^4 - 6x + 7}.$$

(b)

$$g(x) = \sqrt{x} = x^{1/2}.$$

Using the power rule,

$$g'(x) = \frac{1}{2}x^{-1/2}.$$

Thus, for $x > 0$,

$$\boxed{g'(x) = \frac{1}{2\sqrt{x}}}.$$

(c)

$$h(t) = \frac{1}{t} + t^{1/3} = t^{-1} + t^{1/3}.$$

Using the power rule,

$$h'(t) = -t^{-2} + \frac{1}{3}t^{-2/3}.$$

Therefore, for $t \neq 0$,

$$\boxed{h'(t) = -\frac{1}{t^2} + \frac{1}{3}t^{-2/3}}.$$

Math 105 Fall 2026 - Take-Home Quiz 6 Solutions
Due Sunday 10/25/2026 - at 11:55 PM

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(1) Differentiate the following functions. Simplify your answers when possible.

1a.

$$f(x) = \left(\frac{x^2+1}{x^2-1}\right)^{100}.$$

By the chain rule and quotient rule,

$$f'(x) = 100\left(\frac{x^2+1}{x^2-1}\right)^{99} \frac{2x(x^2-1) - 2x(x^2+1)}{(x^2-1)^2}$$
$$= \boxed{-\frac{400x(x^2+1)^{99}}{(x^2-1)^{101}}}.$$

1b.

$$g(x) = \sqrt{8x^4+\sqrt{x}} = \left(8x^4 + x^{1/2}\right)^{1/2}.$$

Thus

$$g'(x) = \frac{1}{2}\left(8x^4+x^{1/2}\right)^{-1/2}\left(32x^3 + \frac{1}{2\sqrt{x}}\right)$$
$$= \boxed{\frac{32x^3 + \frac{1}{2\sqrt{x}}}{2\sqrt{8x^4+\sqrt{x}}}} \qquad (x>0).$$

(2) Let

$$F(x) = (3x-2)^5(5x+1)^6.$$

Using the product and chain rules,

$$F'(x) = 15(3x-2)^4(5x+1)^6 + 30(3x-2)^5(5x+1)^5$$
$$= 15(3x-2)^4(5x+1)^5(11x-3).$$

Therefore $F'(x) = 0$ when

$$3x-2=0, \qquad 5x+1=0, \qquad 11x-3=0,$$

so

$$\boxed{x = \frac{2}{3}, \quad -\frac{1}{5}, \quad \frac{3}{11}}.$$

(3) Let

$$H(x) = f(g(x)), \qquad K(x) = g(x^2-4).$$

For H, the chain rule gives

$$H'(x) = f'(g(x))g'(x).$$

Since $g(1)=2$, $f'(2)=-4$, and $g'(1)=4$,

$$\boxed{H'(1) = (-4)(4) = -16}.$$

For K,

$$K'(x) = g'(x^2-4)(2x).$$

At $x=2$, we have $x^2-4=0$ and $g'(0)=3$, so

$$\boxed{K'(2) = g'(0)(4) = 3\cdot 4 = 12}.$$

MATH 105 Calculus with Algebra – Fall 2026
Midterm 2 Solutions

1. Consider
$$f(x) = 2x^3 + 5x.$$
Then
$$f'(x) = 6x^2 + 5,$$
so the slope at $x = -1$ is
$$f'(-1) = 6 + 5 = 11.$$
Also,
$$f(-1) = 2(-1)^3 + 5(-1) = -7.$$
Thus the tangent line through $(-1, -7)$ is
$$y + 7 = 11(x + 1),$$
or
$$\boxed{y = 11x + 4}.$$

2. For
$$f(x) = \frac{x^2 + 1}{2x + 1},$$
the Quotient Rule gives
$$f'(x) = \frac{(2x)(2x + 1) - 2(x^2 + 1)}{(2x + 1)^2}.$$
Simplifying,
$$f'(x) = \frac{4x^2 + 2x - 2x^2 - 2}{(2x + 1)^2} = \boxed{\frac{2x^2 + 2x - 2}{(2x + 1)^2}}.$$

3. Let
$$f(x) = \sqrt{1 + x^2}.$$
Using the limit definition,
$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{\sqrt{1 + (x + h)^2} - \sqrt{1 + x^2}}{h} \\ &= \lim_{h \to 0} \frac{\sqrt{1 + (x + h)^2} - \sqrt{1 + x^2}}{h} \cdot \frac{\sqrt{1 + (x + h)^2} + \sqrt{1 + x^2}}{\sqrt{1 + (x + h)^2} + \sqrt{1 + x^2}} \\ &= \lim_{h \to 0} \frac{(x + h)^2 - x^2}{h\left(\sqrt{1 + (x + h)^2} + \sqrt{1 + x^2}\right)} \\ &= \lim_{h \to 0} \frac{2x + h}{\sqrt{1 + (x + h)^2} + \sqrt{1 + x^2}} \\ &= \boxed{\frac{x}{\sqrt{1 + x^2}}}. \end{aligned}$$

4. For
$$f(x) = \frac{2x+1}{3x+1},$$
the Quotient Rule gives
$$\boxed{f'(x) = \frac{2(3x+1) - 3(2x+1)}{(3x+1)^2}}.$$

5. Let
$$y = \frac{2x^2 + 3\sqrt{x} + 5}{\sqrt{x}}.$$
Using the Quotient Rule,
$$\boxed{y' = \frac{\left(4x + \dfrac{3}{2\sqrt{x}}\right)\sqrt{x} - (2x^2 + 3\sqrt{x} + 5)\,\dfrac{1}{2\sqrt{x}}}{(\sqrt{x})^2}}.$$

6. For
$$f(x) = \sqrt{3 + (1+x)^4},$$
the Chain Rule gives
$$f'(x) = \frac{1}{2\sqrt{3 + (1+x)^4}} \cdot 4(1+x)^3.$$
Thus
$$\boxed{f'(x) = \frac{2(1+x)^3}{\sqrt{3 + (1+x)^4}}}.$$

7. Rewrite
$$f(x) = \frac{(2+3x)^2}{\sqrt{3-x}} = (2+3x)^2(3-x)^{-1/2}.$$
Using the Product Rule and Chain Rule,
$$\boxed{f'(x) = 6(2+3x)(3-x)^{-1/2} + \frac{1}{2}(2+3x)^2(3-x)^{-3/2}}.$$

8. Let
$$y = (x+1)^2(2x-1)^3.$$
By the Product Rule,
$$\begin{aligned} y' &= 2(x+1)(2x-1)^3 + (x+1)^2 \cdot 3(2x-1)^2 \cdot 2 \\ &= (x+1)(2x-1)^2\big(2(2x-1) + 6(x+1)\big) \\ &= 2(x+1)(2x-1)^2(5x+2). \end{aligned}$$
A horizontal tangent occurs when $y' = 0$. Therefore
$$\boxed{x = -1, \qquad x = \frac{1}{2}, \qquad x = -\frac{2}{5}}.$$

9. The curve is

$$2x^2 + xy = 2 + y^2.$$

Differentiate implicitly:

$$4x + xy' + y = 2yy'.$$

At $(1, 0)$,

$$4 + y' = 0,$$

so

$$y' = -4.$$

The tangent line through $(1, 0)$ is therefore

$$y = -4(x - 1),$$

or

$$\boxed{y = -4x + 4}.$$

10. Let

$$h(x) = x^3 f(x),$$

where $f(2) = 2$ and $f'(2) = 1$. By the Product Rule,

$$h'(x) = 3x^2 f(x) + x^3 f'(x).$$

Thus

$$h'(2) = 3(2)^2 f(2) + (2)^3 f'(2) = 3 \cdot 4 \cdot 2 + 8 \cdot 1 = 24 + 8.$$

Therefore

$$\boxed{h'(2) = 32}.$$

Math 105 Fall 2026 - Take-Home Quiz 7 Solutions

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(1) A particle moves according to

$$s = f(t) = t^3 - 9t^2 + 24t, \qquad t \geq 0.$$

(a) Velocity is the derivative of position:

$$v(t) = s'(t) = 3t^2 - 18t + 24.$$

Factoring,

$$\boxed{v(t) = 3(t-2)(t-4)}.$$

(b) At $t = 1$,

$$v(1) = 3(1)^2 - 18(1) + 24 = 9.$$

Thus

$$\boxed{v(1) = 9 \text{ ft/s}}.$$

(c) The particle is at rest when $v(t) = 0$:

$$3(t-2)(t-4) = 0.$$

Therefore

$$\boxed{t = 2 \text{ s and } t = 4 \text{ s}}.$$

(d) The particle moves in the positive direction when $v(t) > 0$. Since

$$v(t) = 3(t-2)(t-4),$$

the velocity is positive for $t < 2$ and for $t > 4$. Since $t \geq 0$,

$$\boxed{0 \leq t < 2 \quad \text{or} \quad t > 4}.$$

(e) The direction changes at $t = 2$ and $t = 4$, so we find the position at $t = 0, 2, 4, 6$:

$$s(0) = 0,$$
$$s(2) = 2^3 - 9(2)^2 + 24(2) = 20,$$
$$s(4) = 4^3 - 9(4)^2 + 24(4) = 16,$$
$$s(6) = 6^3 - 9(6)^2 + 24(6) = 36.$$

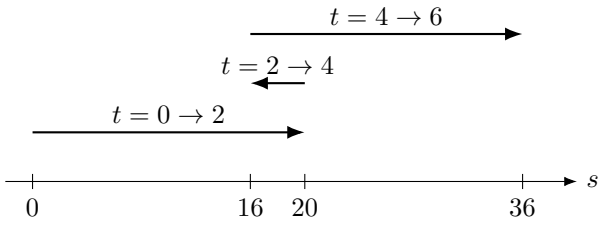
Thus the particle travels from 0 to 20, then back to 16, then forward to 36. The total distance is

$$|20 - 0| + |16 - 20| + |36 - 16| = 20 + 4 + 20 = 44.$$

Therefore

$$\boxed{44 \text{ ft}}.$$

(f) A motion diagram is:



(2) Let L be the length, W the width, and A the area of the rectangle. Then

$$A = LW.$$

Differentiate with respect to t:

$$\frac{dA}{dt} = W\frac{dL}{dt} + L\frac{dW}{dt}.$$

At the instant in question,

$$L = 20, \qquad W = 10, \qquad \frac{dL}{dt} = 8, \qquad \frac{dW}{dt} = 3.$$

Therefore

$$\frac{dA}{dt} = 10(8) + 20(3) = 80 + 60 = 140.$$

Thus

$$\boxed{\frac{dA}{dt} = 140 \text{ cm}^2/\text{s}}.$$

(3) Let x be the distance from the pole to the man, and let y be the length of the man's shadow. The distance from the pole to the tip of the shadow is therefore $x + y$.

By similar triangles,

$$\frac{15}{x+y} = \frac{6}{y}.$$

Cross-multiplying gives

$$15y = 6x + 6y,$$

so

$$9y = 6x \qquad \text{and hence} \qquad y = \frac{2}{3}x.$$

Differentiate:

$$\frac{dy}{dt} = \frac{2}{3}\frac{dx}{dt}.$$

Since the man walks away from the pole at 5 ft/s,

$$\frac{dx}{dt} = 5,$$

so

$$\frac{dy}{dt} = \frac{10}{3} \text{ ft/s}.$$

The tip of the shadow is $x + y$ feet from the pole, so its rate is

$$\frac{d}{dt}(x+y) = \frac{dx}{dt} + \frac{dy}{dt} = 5 + \frac{10}{3} = \frac{25}{3}.$$

Therefore

$$\boxed{\frac{25}{3} \text{ ft/s}}.$$

Notice that the answer does not depend on the man's distance from the pole, so the fact that he is 40 ft from the pole is not needed for the final rate.